

Build two parcel holders

Name: _____ Team: _____ Date: _____

A low tray and an open-sided holder for the same paper parcel.

Gather these materials

- [] Checked base with teacher-approved fixed mounting points
- [] Thin cardboard: two rectangles about 12 cm by 8 cm
- [] Paper parcel about 4 cm by 3 cm by 2 cm
- [] Ruler, pencil, scissors and removable tape

Build: steps 1-4

- [] 1. Fold a small paper parcel about 4 cm long, 3 cm wide and 2 cm high. Tape it closed. Use this same light parcel for every test. Do not add coins or other weights.
- [] 2. Measure the clear space your teacher chose on the base. A 12 by 8 cm card is a starting size. Make it smaller if it would hang over a wheel or cover a control.
- [] 3. For holder A, draw a line 1 cm inside each edge of one rectangle. Cut out the four corner squares. Fold the four sides up along the lines and tape the corners. You have made a low tray.
- [] 4. For holder B, use the second rectangle. Fold its two long edges up by 1 cm and keep both short ends open. Put two folded paper stops near the short ends to stop the parcel sliding out.



Tray plan: cut away the four corner squares.

Fold up the four sides on the dashed lines.

Start with a 1 cm wall. Resize the base to fit.

Drawings show the idea. Measure your own pieces; drawings are not full size.

Finish, code and test

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Build: steps 5-8

[] 5. Turn the robot off. Ask the teacher to show the mounting points. Fix holder A to those fixed points with the approved reusable connectors or paper tape tabs. Do not tape over the motor, sensor or buttons.

[] 6. Put the parcel in holder A. Raise the wheels and check for rubbing. Try the same short drive from lesson 2 three times and record whether the parcel stays on.

[] 7. Turn the power off. Swap to holder B. Use the same parcel, code, floor and start line for three more runs.

[] 8. Choose the better holder using your notes. Change only one feature if neither works well. Keep the selected holder for later lessons.

Make the program

Use lesson 2's checked program.

Low speed, 0.8 seconds forward, then stop and end.

Keep the same settings for both holders.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

Record your test runs

Holder / run	Parcel stayed on?	Any rubbing?	What happened?
A / 1			
A / 2			
A / 3			
B / 1			
B / 2			
B / 3			

A successful build

The holder clears the wheels, sensor and controls.

The parcel stays on in at least two of three runs.

The team records tests for both holders before choosing.

Which holder worked better? Use a result from your notes.